4. Ammonification

Decomposition of organic nitrogen (proteins and amino acids) from dead plants and animals into ammonia is called **ammonification**. Organisim involved in this process are *Bacillus ramosus* and *Bacillus vulgaris*.

5. Denitrification

Nitrates in the soil are converted back into atmospheric nitrogen by a process called **denitrification**. Bacteria involved in this process are *Pseudomonas*, *Thiobacillus* and *Bacillus subtilis*.

$$\text{Nitrate } (NO_3^-) \xrightarrow{\textit{Pseudomonas}} \text{Molecular Nitrogen } (N_2)$$

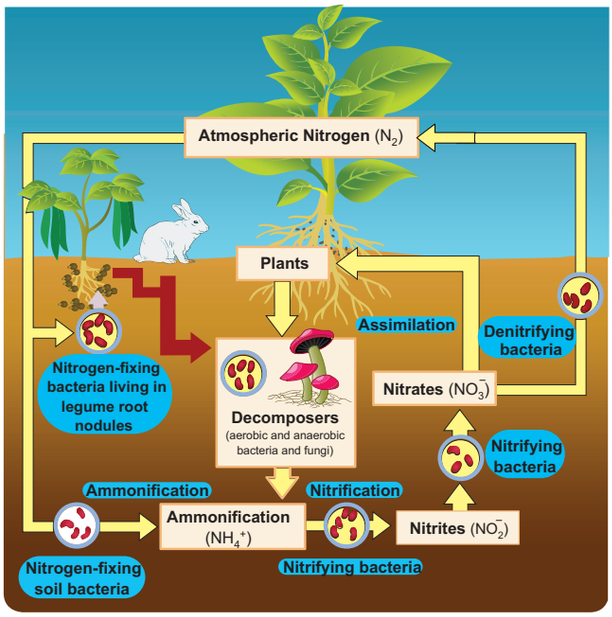


Figure 12.8: Nitrogen Cycle

The overall process of nitrogen cycle is given in Figure 12.8.

12.8.2 Nitrogen Metabolism Ammonium Assimilation (Fate of Ammonia)

Ammonia is converted into amino acids by the following processes:

1. Reductive amination

Glutamic acid or glutamate is formed by reaction of ammonia with α-ketoglutaric acid.

$$\alpha\text{-Ketoglutaric acid} + NH_4 + \xrightarrow[NADPH+H^+ \quad NADP^+]{\text{Glutamate dehydrogenase (GDH)}} \text{Glutamate} + H_2O$$

2. Transamination

Transfer of amino group (NH_3^+) from glutamic acid (glutamate) to keto group of keto acid. Glutamic acid is the main amino acid from which other amino acids are synthesised by transamination. Transamination requires the enzyme transaminase and co enzyme pyridoxal phosphate (derivative of vitamin B6 -pyridoxine)

$$\begin{array}{cccccccc} R1 & & R2 & & R1 & & R2 & \\ | & & | & & | & & | & \\ CH\text{–}NH_3^+ & + & C=O & = & C=O & + & CH\text{–}NH_3^+ & \\ | & & | & & | & & | & \\ COO^- & & COO^- & & COO^- & & COO^- & \\ \text{Amino acid 1} & & \text{Keto acid 2} & & \text{Keto acid 1} & & \text{Amino acid 2} & \end{array}$$

3. Catalytic Amination: (GS/GOGAT Pathway)

Glutamate amino acid combines with ammonia to form the amide glutamine.

$$\text{Glutamate} + NH_4^+ \xrightarrow[ATP \quad ADP + Pi]{\text{Glutamine Synthetase (GS)}} \text{Glutamine.}$$

Glutamine reacts with α ketoglutaric acid to form two molecules of glutamate.

$$\text{Glutamine} + \alpha \text{ Ketoglutaric acid (2- oxoglutarate)} \xrightarrow[NADH+H^+ \quad NAD^+]{\text{GOGAT (enzyme)}} 2 \text{ Glutamate}$$

(GOGAT- Glutamine-2-Oxoglutarate aminotransferase)

12.9 Special modes of nutrition

Nutrition is the process of uptake and utilization of nutrients by living organisms. There are two main types such as **autotrophic** and **heterotrophic** nutrition. Autotrophic nutrition is further divided into **photosynthetic** and **chemosynthetic** nutrition. Heterotrophic nutrition is further divided into saprophytic, parasitic, symbiotic and insectivorous type. In this topic you are going to learn about special mode of nutrition.

12.9.1 Saprophytic mode of nutrition in angiosperms

Saprophytes derive nutrients from dead and decaying matter. Bacteria and fungus are main

XIth_BIO-BOTANY_Ch_12-EM.indd 237 5/17/2020 11:28:06 PM